



## **Pre-attentive Attributes in Graphs Transcript**

Now we will be looking at pre-attentive attributes in graphs. More than text for data visualisation and storytelling, using of pre-attentive attributes in graphs or visual design is of more importance. We have to use them strategically to bring more clarity to our graphics.

Now let us take the example of the bar graphs. The bar graph is a simple design indicating top 10 design concerns and concerns per thousand ordered in a descending fashion. And it can be seen that engine power is less than expected is the top most concern.

And however, there are tyres make excessive noise, engine makes abnormal noise and there is excessive wind noise. There are three noise type complaints. Now the question here is what are the components, what are the noise related complaints? You want to highlight this particular point in the graphic.

The way we can do this is in two forms. One of the pre-attentive attributes to improve the graphic is to use of colour. We can use a specific colour or now we will be looking at pre-attentive attributes in graphs.

The use of pre-attentive attributes is much more relevant for graphs than in text as far as data visualisation is concerned because we will be dealing with graphs and are related visuals to a larger part in our course. We have to pay attention for strategic use of this pre-attentive attribute so that the clarity in our visual improves substantially. For example, look at the graph on the slide.

This shows the car, the reason for car complaints, the reason for top 10 design concerns ordered in the descending order starting with the highest which is the engine power related issues followed by noise related complaints which are categorised into three different categories. Now the same graphic can be improvised by using pre-attentive attribute of colour by highlighting the ones which are above 10% in red colour and the ones in less in grey colour. The management or the product design team can look at the topmost complaints very easily.

The graphic is improving the detail by just using the colour as a pre-attentive attribute. The focus rightly shifts to categories where there are more number of issues. However, this can be further improvised or focused if somebody asks what are the noise related complaints, you can just still use the colour.

However, you can also build in the contrast and get in the light red colour. However, when we want to improve the second graphic, we can use further pre-attentive attributes. We have used colour to improve the graphic.

We can further use contrast as a pre-attentive attribute to drive down the story. The story that we want to present here is what are the noise related complaints. So we

retain the dark red colour for the noise red complaints and use contrast and lighten the background for other complaints or other issues and get more attention to the noise related complaints.

In this way, by using the pre-attentive attribute of colour, you can improve the attention span of the audience and drive home the insight that you want to present in the visual. You will now look at another example of use of pre-attentive attributes in graphs. We will revisit our ticket and response example.

On the right side, on the left side, you have the decluttered tickets issued and tickets responded graphic, which was clean and decluttered and made much more responsive. On the right hand side, we have more pre-attentive attributes added to this graphic to make it more revealing and insight driven. If you remember, the insight for the graphic was to make a decision about hiring additional people.

Now we are trying to build in that insight by using pre-attentive attributes. We are now adding labels to the complaints that are to the tickets that are received and the complaints and the tickets that are processed so that the productivity dip can be easily seen and measured. So a decision can be taken.

The same graphic with the same logic of labels being right next to the series, being less cluttered and adding little amount of labels is driving more and better insight as compared to the plain graph. Though it is decluttered, the context is improvised because you are trying to answer the question whether you need to hire more people to address the fall in productivity gap. So you are just getting more labels added to the series and that will help in making the decision even more faster.

So this is the use of pre-attentive attributes in a graph that can help the visual become more effective in delivering the message. We will now see the use of size as a pre-attentive attribute. Size is proportionally including size indicates the importance proportionally.

Larger is the size of an element on a graphic, higher is its importance, at least the perceived importance. Therefore, projecting size of an element should really reflect the importance in the design that we want to communicate. Otherwise, we will be inflating or deflating the actual importance of the attribute.

Keeping this in mind, let me drive you to an example or the real-life case that has happened. In Google, we will now see the use of size as a pre-attentive attribute. Size communicates importance.

Larger elements are naturally given higher importance in the visuals. Equal importance represents equal sizing or presumes equal sizing. Therefore, when we relatively increase the size of an element in the visual, it implies that we are assigning higher importance.



One has to be very careful regarding the sizing of the elements on the visual space or on the graph. Because if we disproportionately increase or decrease the size of an element, we will be forcing an intuition on the audience to assign higher or lower importance. Sometimes for lack of data, it is very eager or we are desperate to increase the size to show, to fill the graphic, to create an impression that the data is more important.

This should be avoided at all cost because at times it might lead to wrong decisioning and derails the entire purpose of building a credible story. And in cases where you cannot use the size, just leave it simple so that the audience can take their own interpretations.

Thank you

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